# DeliverCheck — Email Deliverability & Bounce Verifier

Project Submission & Architectural Overview Document

| PROJECT DEPLOYMENT LINKS | STATUS & DETAILS |
|---|---|
| **Public Live Application:** https://across-recipients-examining-patrick.trycloudflare.com | ● **Live & Accessible Worldwide** Full interactive web application |
| **GitHub Repository:** https://github.com/Thanusrithika/email-deliverability-checker | **Source Code & README** Includes Docker, Render & Railway configs |
| **Live Health Diagnostics API:** https://across-recipients-examining-patrick.trycloudflare.com/api/health | **REST JSON Endpoint** Returns server health & port 25 status |

## 1. Executive Summary & Core Objective

DeliverCheck is a web application and REST service that analyzes a list of email addresses and classifies each as **Likely to be delivered**, **Likely to bounce**, or **Unknown**. Users can paste multiple addresses in arbitrary formats (newline-delimited, comma-separated, space-separated, or CSV) and immediately view diagnostic metrics, filter results, inspect RFC-level breakdowns, and export reports.

## 2. Deliverability Classification Logic

| Category | Definition | Trigger Conditions |
|---|---|---|
| **Likely to be delivered** | Recipient mailbox exists and will accept incoming messages. | Valid RFC syntax, active MX records, destination server returns SMTP 250 OK for RCPT TO, domain is not Catch-All. |
| **Likely to bounce** | Email delivery will fail with certainty. | Malformed syntax, consecutive dots, non-existent domain (NXDOMAIN), Null MX (RFC 7505), or permanent mailbox rejection (SMTP 550/551/553). |
| **Unknown** | Mailbox existence cannot be definitively confirmed without sending a message. | Catch-All mail server (accepts all addresses indiscriminately), greylisting/temporary deferral (SMTP 421/450/451), or outbound port 25 blocked by host. |

## 3. Mail Server Behaviors & Cloud Hosting Restrictions

**Why 'Unknown' is a valid result:** Many modern mail servers (Google Workspace, Microsoft 365) employ Catch-All policies to prevent directory harvesting: they return 250 OK for every address regardless of whether the user exists, then discard or bounce later. Because no handshake can differentiate an existing from a non-existent mailbox, **Unknown is the only**

**accurate classification**.

**Cloud Hosting Outbound Port 25 Blocking:** Free and shared cloud hosts (Render, Railway, PythonAnywhere, AWS, DigitalOcean) block outbound TCP port 25 to protect against spam bots.
• **What the deployed version checks:** Full RFC 5322 syntax validation, DNS MX queries, RFC 7505 Null MX detection, NXDOMAIN detection, disposable provider detection, and role account tagging. Any syntax/domain failure is flagged as Likely to bounce.
• **Safe Fallback:** When port 25 is restricted, mailbox verification defaults safely to 'Unknown' with explicit diagnostic notes.
• **Compliance:** The tool never sends actual email messages; sessions cleanly terminate with QUIT.

## 4. How to Run the Tool Locally

```
# 1. Clone repository
git clone https://github.com/Thanusrithika/email-deliverability-checker.git
cd email-deliverability-checker

# 2. Setup Virtual Environment
python -m venv venv
venv\Scripts\activate      # Windows (or source venv/bin/activate on Linux)

# 3. Install Dependencies & Run Tests
pip install -r requirements.txt
python test_verifier.py

# 4. Launch Application Server
python app.py               # Open http://localhost:5000
```